

# Hamza Ammar

Waterloo, Ontario — Email — LinkedIn — GitHub

## EDUCATION

### University of Waterloo

Bachelor of Software Engineering (BSE)

Elected Class Academic Representative

## EXPERIENCE

### Cohere

Intern of Technical Staff

Toronto, ON  
Incoming Jan 2027

- Incoming on the **Forward Deployed Engineering** team, building agentic platform tooling

### Shopify

Software Engineering Intern

Toronto, ON  
May 2026 - Present

- Built **25+** MCP agent tools on **Rails/OAuth APIs** for AI-assisted marketing operations
- Designed **write-preview safety gates** for high-risk **mutating agent workflows** before AI edits execute
- Engineered **BigTable-backed push pacing** for notifications serving **500K+ daily merchant recipients**
- Hardened **4-stage recipient-list imports** resumable to recover failed backfills without full restarts

### Dealish

Founding Engineer

Waterloo, ON  
Dec 2025 - Present

- Built and launched **React Native (Expo)** location-based app for **real-time** food and drink discovery
- Designed a secure **Supabase backend** with **role-based access control** to manage user profiles and deals
- Integrated a **Google Sheets inventory** sync via OAuth for partner restaurants to **manage deals** in real-time

### Kuzu Inc

Software Intern

Waterloo, ON  
Jun 2024 - Aug 2024

- Built **operator printing** for KuzuDB's query planner, improving plan visualization and **cutting debugging time**
- Added metadata printing for **40+ operators**, **increasing clarity** of query plans used in performance benchmarking
- Improved Explorer design by refining node layouts and using **Chroma.js** pastel colours for a **cleaner interface**

## PROJECTS

### Horizon MCP

January 2026 - May 2026

- Built a **multi-tenant MCP server** on **AWS ECS Fargate** serving course data to **50+ university students**
- Encrypted user credentials with **AWS KMS** and session state via **AES-256-GCM envelope encryption**
- Built **pgvector semantic search** over notes and Piazza using **HNSW-indexed** OpenAI embeddings
- Integrated **D2L, Piazza, and Crowdmark**, handling **Duo 2FA** through per-user **noVNC** browser sessions

### Godseye

March 2026

- Built a **multi-agent** prediction market tool using **AI-driven simulations** on Polymarket bets
- Placed **2nd** in the Polymarket track at **YHack (Yale)**
- Designed **agent orchestration pipeline** where **specialized agents** debate and produce consensus trading signal

### Deep Learning Chess Engine

November 2025

- Developed a **deep CNN-based chess engine** trained end-to-end on millions of grandmaster games
- Implemented a **residual convolutional architecture** with policy and value heads using **PyTorch**
- Optimized training on **NVIDIA H100s** via efficient tensor operations, large batch sizes, and mixed precision

### CourseConnect

Sep 2025 - Dec 2025

- Built a **Python + Playwright scraper** to collect and normalize **1,000+** courses into a structured dataset
- Constructed a **prerequisite dependency graph** enabling SE students to **validate and plan degree paths**

## CERTIFICATIONS

- Machine Learning Specialization - *DeepLearning.ai, Stanford University* Aug 2025
- Algorithmic Toolbox - *UC San Diego* Jan 2025

## SKILLS

- **Languages:** Python, C++, C, Go, TypeScript, JavaScript, SQL, Ruby, Rust
- **Frameworks & Libraries:** React, Next.js, Node.js, Express.js, PyTorch, TensorFlow, Playwright, Flask
- **Cloud & Infra:** AWS (ECS, S3, KMS), Docker, Supabase, PostgreSQL, pgvector, Prometheus, Git/GitHub
- **Technical Areas:** Full-Stack Development, Machine Learning, Distributed Systems, Algorithms & Data Structures